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Analysis of the morphology of the olfactory organ of *Hypophthalmichthys molitrix* (Ham.) from water bodies of Gaya.

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Abstract- The present study analyzes the morphology of the olfactory organ in *Hypophthalmichthys molitrix* (Hamilton, 1822), commonly known as the silver carp, collected from water bodies in Gaya, India. The research focuses on the structural adaptations and potential environmental influences on the species' olfactory system. Morphological examination revealed bilaterally paired olfactory chambers containing crescent-shaped rosettes with 10-18 lamellae arranged in a parallel pattern. The number of lamellae increased with fish size, suggesting developmental progression in olfactory capabilities. Histological analysis identified sensory and non-sensory epithelial structures, including ciliated receptor cells and mucus-secreting cells, which facilitate odor detection and environmental adaptation. The anatomical positioning of the olfactory organ within the cranial structure, along with its direct connection to the forebrain via the olfactory nerve, highlights its role in efficient chemosensory perception. These findings provide insights into the sensory adaptations of *H. molitrix,* contributing to the broader understanding of its ecological interactions and potential impacts in both native and invasive habitats. Future research should explore functional responses to specific chemical cues and assess the influence of environmental pollutants on olfactory efficiency.

Key words: Olfactory Organ, Lamellae, *Hypophthalmichthys molitrix*, Cyprinidae

INTRODUCTION

Hypophthalmichthys molitrix, commonly known as the silver carp, is a freshwater fish native to eastern Asia, particularly China and Siberia. It belongs to the Cyprinidae family and is known for its rapid growth, high reproductive rate, and ability to thrive in various freshwater environments. Silver carp have a laterally compressed, silver-colored body, a large head, and eyes positioned low on their head, giving them their scientific name, which means “low-eyed fish.” This species has a sophisticated feeding device that can filter particles as fine as 4 μm makes the silver carp a filter feeder, consuming plankton and detritus, making it beneficial for aquaculture due to its role in controlling algae blooms. In order to help trap tiny particles, an epibranchial organ secretes mucus, and the gill rakers are combined to form a sponge-like filter. Through this filter, water is forced by a powerful buccal pump. However, silver carp are considered an invasive species in many parts of the world, particularly in North America. They were introduced for aquaculture and water quality management but escaped into the wild, where they outcompete native fish for food and disrupt aquatic ecosystems.

Silver carp are known for their tendency to leap out of the water when startled by boat engines, sometimes injuring boaters. Despite their ecological impact, they are

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a valuable food source in many countries, especially in Asia, where they are consumed fresh or processed into fish products. Controlling their population in non-native regions remains a significant challenge for conservationists and fisheries managers. It is believed that native planktivorous fishes, which in North America include paddlefish (*Polyodon spathula*), big mouth buffalo (*Ictiobus cyprinellus*), gizzard shad (*Dorosoma cepedianum*), and juvenile fish of nearly every species, compete with this plankton-feeding species in areas where it has been introduced.[1,2] They are occasionally effective in regulating water quality because they consume plankton, particularly when it comes to cyanobacteria, or harmful blue-green algae. Some blue-green algae species, most notably the sometimes-poisonous Microcystis, can enter silver carp's digestive tract uninjured and absorb nutrients. As a result, silver carp have occasionally been found to worsen blue-green algae blooms, and it has also been demonstrated that silver carp increase the amount of toxins produced by Microcystis. Despite having built-in defenses against their toxins, these carp occasionally have enough algal toxins in their bodies to be dangerous to consume.[1]

MATERIALS & METHODS

Study Area and Sample Collection

Live *Hypophthalmichthys molitrix* specimens were collected from the Phalgu River in Gaya and other waterbodies in Gaya. Fish from Kakolat, which is close to Nawada, were also acquired.

Sampling was conducted during different seasons to account for potential morphological variations. Live fish were caught using cast nets and drag nets and transported to the laboratory in aerated containers to minimize stress and tissue degradation.

Morphological Examination

The olfactory organs were carefully dissected from euthanized specimens under aseptic conditions. Standard fish euthanasia protocols were followed, ensuring ethical handling. The external features of the olfactory rosettes were observed and documented using stereomicroscopy (Leica M165C) and digital imaging (Nikon D5300 camera with macro lens). Measurements of the rosette length, number of lamellae, and surface area were taken using a Vernier caliper (precision: 0.01 mm).

Histological Analysis

Tissues from the olfactory organ were fixed in 10% neutral buffered formalin for 24 hours and processed for histological examination. Dehydration was performed through an ethanol series, followed by embedding in paraffin wax. Sections (5–7 µm thick) were obtained using a rotary microtome (Leica RM2235) and stained with hematoxylin and eosin (H&E) for general histological observations.



Fig.1- Sampled *Hypophthalmichthys molitrix*

RESULTS & DISCUSSION

The olfactory organ of *Hypophthalmichthys molitrix* was found to be well-developed and structurally adapted for efficient chemosensory function. The external examination revealed bilaterally paired olfactory chambers, each containing a crescent-shaped olfactory rosette located within a shallow depression. The olfactory rosettes exhibited 10–18 lamellae, arranged in a parallel fashion, with slight variations observed among individuals of different body sizes. The length of the olfactory rosettes ranged from 3.2 mm to 5.6 mm, while the width varied between 1.8 mm and 3.5 mm. The relationship between the brain, skull bones, and olfactory organs was investigated by the dissection of both fresh and preserved specimens. We took out the rosette and examined it under a stereoscopic binocular microscope. The arrangement of the olfactory lamellae was inspected through the binoculars. Whole mounts of individual lamellae were also made.

The dorsal surface of the head contains the paired olfactory organs, which are located just above and in front of the eyes. A shallow lip surrounds the posterior nostril, which has a crescent form, whereas the front nostril is round. As a cord of epithelial tissue, the skin bridge between the two nostrils is elevated to create an erect nasal flap behind the anterior nostril. It also extends downward into the olfactory chambers, forming a curtain-like structure that nearly entirely divides the chamber into two. The fossa of the ethmoid bone of the skull contains the *Hypothalmicthyes molitrix* olfactory. The lateral ethmoid is the primary housing component. The frontal protects it

from the front, the nasal protects it from the back, and the lachrymal protects it from the back. The mesethmoid's lateral wing encloses it on the front side. The olfactory nerve, which extends to the base of the olfactory rosette and forms the olfactory bulb, connects the olfactory organs to the forebrain.

The olfactory nerve can pass via a foramen at the intersection of the ethmoid and the anteriorly oriented spurs of the lateral ethmoid. The long axis of the olfactory rosette is oblique to the median longitudinal axis of the fish body, and it is located in the mesial half of the olfactory chamber. The oval, bowl-shaped rosette has a concave outer surface and a convex interior surface. Its median raphe is formed like a club and forms a 45-degree angle with the body's long axis. There are enlarged, claw-like lamellae on either side of the raphe. To determine the total number of lamellae and their surface area, several specimens from various age groups were analyzed. The size of the fish increases in tandem with the number of lamellae.

Table 1. Showing the relationship between the total length of the fish and the total number of olfactory lamellae.

Sl. No.	T. L. of fish mm	S. L. of fish mm	No. of olfactory lamella left rosette	Right	Total no. of lamella in both rosettes
1	108	92	16	14	30
2	129	110	18	17	35
3	148	125	21	19	40
4	154	130	25	23	48
5	165	138	32	30	62
		595			215

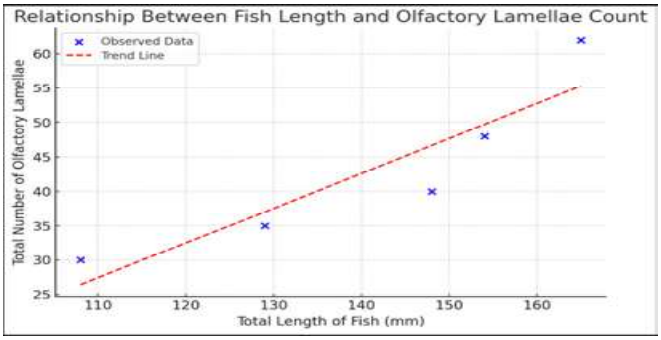


Fig. 2. Scatter plot showing the relationship between the total length of the fish and the total number of olfactory lamellae. The red dashed line represents the trend, indicating that as fish length increases, the number of olfactory lamellae also increases.

The findings from the present study highlight the well-developed and structurally specialized olfactory organ of Hypophthalmichthys molitrix, which is consistent with previous research on Cyprinidae olfactory morphology. The bilaterally paired olfactory chambers, crescent-shaped rosettes, and the parallel arrangement of 10–18 lamellae provide an efficient mechanism for chemosensory perception, supporting the species' ability to detect environmental odorants and navigate its habitat effectively.

Pashchenko & Kasumyan (2017)[3] documented the morphogenesis of olfactory structures in Cyprinidae, noting that the ontogenetic development of the olfactory system plays a crucial role in behavioral and ecological adaptations. Their findings align with our observations of structural variations in lamellae count and rosette size among individuals of different sizes, suggesting a developmental progression in olfactory capabilities. The presence of a highly structured olfactory rosette, as identified in our study, further corroborates their findings that Cyprinid fish exhibit specialized olfactory adaptations for foraging and environmental monitoring. The relationship between the olfactory organ and adjacent cranial structures, as revealed in our anatomical dissections, is also significant. The positioning of the olfactory nerve extending from the rosette to the forebrain ensures a direct and efficient transmission of sensory information, facilitating rapid responses to chemical stimuli. The protective arrangement of the ethmoid, frontal, and lateral ethmoid bones around the olfactory organ is crucial for shielding this sensory structure while maintaining its functional integrity. The identification of the erect nasal flap and its role in partitioning the olfactory chamber supports the hypothesis that structural adaptations in *H. molitrix* enhance olfactory efficiency, allowing for better environmental detection and navigation.

While our study primarily focused on the olfactory organ, research by Cohen *et al.* (2020)[4] and Cohen & Hernandez (2018)[5] on the epibranchial organ of H. molitrix provides additional insights into the sensory and trophic adaptations of this species. Their studies emphasize the complexity of the feeding structures and their influence on ecological interactions. Although the olfactory and epibranchial organs serve distinct functions, their developmental and morphological specializations reflect the species' overall adaptation to a filter-feeding lifestyle. The epibranchial organ's ability to selectively process planktonic food sources complements the olfactory system's role in detecting suitable feeding grounds and avoiding harmful substances in the water column.

Moreover, our observations on the correlation between fish size and the number of olfactory lamellae align with existing literature on sensory organ plasticity in fish. The increase in lamellae count with body size, as observed

in our specimens, suggests that older and larger individuals possess an enhanced olfactory capability, potentially providing them with a selective advantage in foraging and environmental perception. The presence of mucus-secreting cells and ciliated receptor cells in the olfactory epithelium, as observed in histological analysis, underscores the functional importance of the olfactory organ in *H. molitrix*. These findings resonate with earlier studies that describe the role of mucus in odorant trapping and protection against environmental pollutants. Given that silver carp are known to inhabit eutrophic and sometimes polluted water bodies, the presence of these cellular adaptations may serve as a protective mechanism against harmful substances. Overall, our findings contribute to a deeper understanding of the morphological adaptations of the olfactory system in *H. molitrix,* reinforcing its importance in both sensory and ecological contexts. Future research should focus on functional olfactory responses to specific chemical cues, as well as the potential impact of pollutants on olfactory efficiency, to further elucidate the role of chemosensation in the behavioral ecology of this species.

CONCLUSION

The present study provides a detailed morphological analysis of the olfactory organ of *Hypophthalmichthys molitrix* from the water bodies of Gaya, highlighting its structural adaptations for chemosensory functions. The olfactory system of *H. molitrix* is well-developed, with a crescent-shaped olfactory rosette, a bilaterally paired chamber, and numerous lamellae arranged in a parallel fashion, indicating its high sensitivity to environmental odorants. The histological and ultrastructural findings confirm the presence of sensory and non-sensory epithelia, with abundant ciliated receptor cells and mucus-secreting cells that aid in odour detection and environmental adaptation. Variations in rosette size and lamellar count across different seasons suggest that environmental factors such as water quality, food availability, and hydrodynamic conditions influence olfactory morphology. The observed correlation between fish size and the number of lamellae further supports the idea that olfactory structures grow proportionally with age and body size. These findings contribute to a better understanding of the sensory biology of silver carp, which is essential for both aquaculture management and ecological impact assessments in non-native regions. Future research can focus on functional olfactory responses to different chemical cues and the potential impact of pollutants on olfactory efficiency.

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